ModelIt!

Teacher's Guide

Why Do Roller Coasters Make You Scream?

Modeling Forces, Energy Conversion, and the Physics of Thrills

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-PS2-1, HS-PS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS2-1, HS-PS3-2

Analyze data to support the claim that Newton's second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motion of particles and energy associated with the relative positions of particles.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical models (PE=mgh, KE=½mv^2, F=ma, ac=v^2/r) to predict coaster velocity, G-forces, and energy distribution at every point on the track.
Disciplinary Core Idea	**PS2.A Forces and Motion / PS3.A Definitions of Energy** — Newton's second law describes the relationship between net force, mass, and acceleration. Energy at the macroscopic scale can be accounted for as combinations of kinetic and potential energy.
Crosscutting Concept	**Energy and Matter** — Students track the conservation and conversion of energy throughout the coaster system, accounting for potential, kinetic, and thermal (friction) energy at every point.

CAST Testing Alignment

MULTIPLE CHOICE:

A roller coaster with a first hill height of 50 meters reaches the bottom of the drop. Ignoring friction, which expression best predicts the coaster's velocity at the bottom?

CONSTRUCTED RESPONSE:

Using your model and energy conservation principles, explain why the second hill on a roller coaster must always be shorter than the first. Include the role of friction in your explanation and calculate the maximum height of the second hill if the first hill is 60 meters and 10% of energy is lost to friction.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Roller coasters are physics textbooks made of steel. Every drop, loop, and corkscrew is an exercise in energy conservation, Newton's laws, and human biomechanics. Engineers don't just build exciting rides — they calculate exactly how much force the human body can handle and design tracks that push right up to that limit without crossing it.

Energy Conservation on a Coaster

The chain lift at the start is the coaster's only energy input. It converts electrical energy into gravitational potential energy ($PE = mgh$). From the top of the first hill, all motion comes from converting this stored potential energy into kinetic energy ($KE = ½mv^2$). At any point, PE + KE + Energy lost to friction = Initial PE. This is why every subsequent hill must be shorter — friction continuously removes energy from the system.

Why Mass Doesn't Matter

Setting PE = KE: $mgh = ½mv^2$. The mass cancels: $v = \sqrt{(2gh)}$. This means a 1,000 kg coaster and a 10,000 kg coaster reach the SAME speed at the bottom of the same drop. This is Galileo's insight — all objects fall at the same rate regardless of mass. In practice, heavier coasters lose slightly less speed to air resistance (proportionally) but more to friction.

G-Forces and Human Limits

G-force is the acceleration experienced relative to gravity. At the bottom of a drop, centripetal acceleration adds to gravity: riders feel 1G (gravity) + v^2/rg additional Gs. In a 3G turn, a 70 kg person feels 210 kg of force. Fighter pilots black out above 9G; most people become uncomfortable above 4G sustained. Coaster designers keep instantaneous peaks below 6G and sustained forces below 4G.

The Art of Airtime

Airtime — the feeling of weightlessness — occurs when the coaster crests a hill and the track curves away faster than gravity pulls riders down. At the top of a parabolic hill, riders experience less than 1G — even 0G or negative G (floating off the seat). Engineers calculate the exact hill shape needed for each airtime moment using $v^2/r = g$ for 0G. This is what makes riders scream with delight.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify coaster mass, track height, velocity, G-force on rider, and energy conversion as the key components of the roller coaster physics system.

E **Establish Relationships**
Students determine that track height sets the potential energy budget, velocity depends on how much potential energy has converted to kinetic, and G-force depends on velocity and track curvature.

V **Visualize & Model**
Students build a computational model showing energy conversion along the track and G-force at key points including drops, loops, and hills.

E **Evaluate Outcomes**
Students run maximum drop, mass comparison, and loop-the-loop scenarios to verify energy conservation and discover the relationship between track geometry and rider experience.

R **Revise & Extend**
Students add friction/air resistance, track curvature radius, or rider body position to model more realistic coaster dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The fastest coasters hit 150 mph. The most extreme pull 6 Gs — close to what fighter pilots experience. How do engineers build a ride that terrifies you but never hurts you?"*

Watch For

Show a POV roller coaster video for 15 seconds. Ask: What's happening to your ENERGY and your BODY?

Slide 2: Learning Objectives

3 min

Say: *"Today we're using physics to design the most thrilling ride possible — and proving it's safe with math."*

Watch For

Have students read objectives. Pre-teach 'gravitational potential energy' and 'G-force.'

Slide 3: Big Question

3 min

Say: *"At the top of the first hill, you're barely moving. At the bottom, you're going 100 mph. Where did that energy come from?"*

Watch For

Students draw energy bar charts for top-of-hill vs. bottom-of-hill. Compare predictions.

Slide 4: LEVER Framework

2 min

Say: *"We're building a physics model that can calculate the speed and G-force at every point on a roller coaster."*

Watch For

Preview LEVER steps. Emphasize this is engineering design backed by physics calculations.

Slide 5: Activity 1: Components

8 min

Say: *"What can coaster designers CHOOSE? What do the laws of physics DETERMINE for them?"*

Watch For

Guide sorting. Mass and height are designer choices; velocity, G-force, and energy conversion are physics outcomes.

Slide 6: Activity 2: Connections

8 min

Say: *"The first hill is your energy bank. Every meter of height is money in the account. Every meter of friction is money spent."*

Watch For

Introduce PE=mgh, KE=½mv^2, and show how conservation connects height to speed.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's race — heavy coaster vs. light coaster. Same track. Who gets to the bottom first?"*

Watch For

Students predict mass effects. Run simulation. Discover mass cancels. Then test the loop.

Slide 8: Discoveries

5 min

Say: *"Mass doesn't matter. Galileo was right 400 years ago. And now you know why the second hill is always shorter."*

Watch For

Connect discoveries to energy conservation. Discuss where friction fits in the real world.

Slide 9: STEM Challenge

12 min

Say: *"The theme park wants your design. 60 meters max height. At least one loop. Prove with math that nobody gets hurt."*

Watch For

Groups design tracks, calculate velocities and G-forces at key points. Present and defend designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Coaster Mass, Track Height
- **INTERNAL:** Velocity, G-Force on Rider, Energy Conversion

Relationships

- **Track Height to Velocity:** POSITIVE (+) — Greater track height means more gravitational potential energy at the top. As the coaster drops, this converts to kinetic energy: $v = \sqrt{(2gh)}$. A 60m drop produces about 34 m/s (76 mph) at the bottom, ignoring friction.
- **Velocity to G-Force on Rider:** POSITIVE (+) — Higher velocity through curves and loops produces greater centripetal acceleration ($a = v^2/r$), which translates to higher G-forces felt by riders. Doubling velocity quadruples the centripetal force.
- **Track Height to Energy Conversion:** DETERMINES — The first hill height sets the total energy budget for the entire ride. Energy Conversion at any point follows conservation: KE + PE + friction losses = initial PE. No subsequent hill can exceed the first without additional energy input.

Key Discoveries

- The first hill of a roller coaster is its entire energy budget — every subsequent hill must be shorter because friction continuously removes energy from the system
- Coaster mass does NOT affect the speed at the bottom in ideal conditions (just like Galileo's famous experiment) — the velocity depends only on the height dropped
- G-force is created by centripetal acceleration in curves and loops — riders feel heaviest at the bottom of drops and lightest (or weightless) at the top of hills
- Energy conservation means the coaster can never go higher than its first hill without additional energy input — the chain lift at the start is the only motor

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavier coasters go faster"	Mass cancels from the energy conservation equation: $v = \sqrt{2gh}$. A heavier coaster has MORE energy but also requires MORE energy to accelerate — the effects cancel perfectly. This is Galileo's insight: all objects fall at the same rate regardless of mass (in a vacuum). In practice, heavier coasters lose slightly less speed to air resistance per unit mass, but this is a small effect.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The motor drives the coaster through the whole ride"	The chain lift or launch at the start is the coaster's ONLY energy input. After that, the ride runs entirely on gravity and energy conservation. Every drop, loop, and turn uses energy from the first hill. The brakes at the end REMOVE energy — they don't add it. This is why the ride always comes back to a lower point than where it started.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"G-forces are dangerous — the more the better for thrills"	G-forces are only thrilling within a narrow range. Below 0.5G, riders feel pleasantly floaty. At 3-5G, the sensation is intense and exciting. Above 6G sustained, blood pools in the legs and riders can lose vision (greyout) or consciousness (blackout). Above 9G, even fighter pilots in pressure suits can lose consciousness. Engineers design for the 'sweet spot' — intense enough to thrill, safe enough to ride 100 times.